

Valentina Montaña Sandez

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EDUCATION

Universidad Jesuita de Guadalajara (ITESO)

Bachelor of Science in Computer Systems Engineering, GPA 8.8/10

Guadalajara, Jalisco

Aug. 2021 - Dec. 2026

EXPERIENCE

Software & Data Analytics Intern

Guadalajara, Jalisco

ITISA

May 2026 - Present

- Develop Python/FastAPI services that receive and process people-counting analytics events from Avigilon systems through REST endpoints, supporting ENTRY/EXIT metrics and operational reporting.
- Design local data persistence and monitoring workflows with SQLite and CSV for isolated Windows/NVR environments with limited internet connectivity.
- Develop a dashboard that consolidates availability, downtime, and log data across three network video recorders for faster operational review.
- Support an airport alarm-monitoring application integrated with Avigilon, including API connectivity, Windows deployment, firewall/port validation, and repeatable launcher procedures.

Cloud Operation Intern

Guadalajara, Jalisco

C&A Mexico

March 2025 - March 2026

- Designed, deployed, and supported cloud environments across Microsoft Azure and OCI, including virtual machines, private networks, subnets, security rules, and storage services.
- Automated recurring administrative and operational tasks with PowerShell and supported reporting/data workflows using Power BI, SSAS, and KNIME.
- Implemented Azure DevOps CI/CD workflows and secure connectivity between build agents and OCI environments for application deployments.
- Deployed containerized .NET 8 and Python services in OCI; configured load balancers, SSL/TLS certificates, HTTPS listeners, and backend routing.
- Performed monitoring, backup/restore validation, incident troubleshooting, architecture documentation, and cloud cost analysis.

PROJECTS

Doble Punto - Modular Multi-Tenant Point-of-Sale Platform

2026 - Present

- Co-founded and currently develop a full-stack POS product for small businesses, covering sales, mixed payments, cash-register shifts, customers, inventory, analytics, and role-based access.
- Built an Apple Wallet-inspired digital loyalty-card experience with unique member codes, automatic purchase stamps, five-purchase reward generation, and auditable \$0 reward redemptions linked to the customer and cash shift.
- Developed the platform with TypeScript, React 19, Next/Vinext, Cloudflare D1/SQLite, Drizzle ORM, R2 object storage, and Cloudflare Workers, including authenticated REST endpoints and incremental database migrations.
- Defined a modular multi-tenant architecture with server-side authorization, data isolation, auditability, and operation rules designed for atomicity and idempotency.

Data Warehouse & Business Intelligence Solution

2026 - In Progress

- Designing a SQL Server dimensional warehouse from Northwind using fact and dimension tables, date/surrogate keys, joins, and star-schema modeling.
- Building stored procedures for incremental and idempotent loads while validating duplicates, keys, data types, and collation consistency.
- Preparing analytical measures and a Power BI/Tableau dashboard to visualize business indicators from the warehouse model.

AWS Image Recognition & MLOps Pipeline

2026

- Contributed to a Python access-control prototype using AWS Rekognition for face comparison and liveness validation, S3 for image storage, and PostgreSQL/RDS for application data.
- Applied AWS-based pipeline design, CI/CD, model/data versioning, inference-code and model-artifact packaging, testing with pytest, and experiment tracking with MLflow.

PROJECTS - CONTINUED

Polyglot IoT Data Platform - NoSQL Systems

Academic Project

- Modeled an IoT data platform across MongoDB, Dgraph, and Cassandra for device metadata, graph relationships, and time-series sensor readings.
- Implemented indexes, aggregations, lookups, graph queries, and partition-key strategies for device, alert, log, and administrative-event use cases.

Real-Time Metropolitan Traffic Monitor - AWS Cloud Architecture

Academic Project

- Designed an event-driven architecture using AWS Lambda, S3, EC2, and Route 53 for ingestion, processing, and visualization of urban traffic data.
- Applied distributed-cloud concepts for availability, fault tolerance, and lower-latency processing.

TECHNICAL SKILLS

Data Engineering & BI: SQL, SQL Server, PostgreSQL, SQLite, dimensional modeling, star schemas, ETL/ELT, data validation, Power BI, Pandas, NumPy, KNIME

Programming & APIs: Python, FastAPI, REST APIs, TypeScript, JavaScript/Node.js, React, Next.js, PowerShell, Bash, HTML/CSS

Cloud & DevOps: AWS, Microsoft Azure, Oracle Cloud Infrastructure, Cloudflare Workers/R2, Docker, Azure DevOps, Git/GitHub, CI/CD

MLOps & Testing: AWS Rekognition, MLflow, model/data versioning, inference packaging, Makefiles, linting, pytest

Systems & Databases: Linux, Windows Server, Cloudflare D1, Drizzle ORM, MongoDB, Cassandra

Languages: Spanish - Native; English - Intermediate (B1-B2)